

## d-Block transition elements (group 3-12)

Electronic configuration :  $(n-1)d^{1-10} ns^{1-2}$

Exceptions  $\left\{ \begin{array}{l} \text{Cr} = 4s^1 3d^5 \\ \text{Cu} = 4s^1 3d^{10}, \text{Pd} = 5s^0 4d^{10} \end{array} \right.$

Non-typical transition elements  
Zn, Cd & Hg

Physical properties:

-High melting and boiling point

Melting point : s-block metals < d-block metals

Sc < Ti < V < Cr > Mn < Fe > Co > Ni > Cu > Zn (3d Series)

Melting point  $\left\{ \begin{array}{l} \text{Zn} > \text{Cd} > \text{Hg} \\ \text{Cu} > \text{Ag} < \text{Au} \end{array} \right.$

-High enthalpies of atomization (Highest for V in 3d, series)

Atomic Radius:

3d series: Sc > Ti > V > Cr > Mn > Fe = Co = Ni < Cu < Zn

In a group 3d < 4d = 5d (Lanthanide contraction)

eg: Ti < Zr = Hf  $\left\{ \begin{array}{l} \text{Smallest radius} - \text{Ni} \\ \text{Largest radius} - \text{La} \end{array} \right.$

Density:

s-Block < d-Block

3d series: Sc < Ti < V < Cr < Mn < Fe < Co < Ni < Cu > Zn

In a group 3d < 4d < 5d

- Ionisation enthalpy: increases from left to right
- Oxidation states : Variable; higher O.S. stable down the group
- Trends in  $E^\circ_{M^{2+}/M}$  :  $E^\circ$  for Mn, Ni and Zn are more negative than expected.
- Trends in  $E^\circ_{M^{3+}/M}$  : variable
- Chemical reactivity and  $E^\circ$  values : Variable  $\text{Ti}^{2+}$ ,  $\text{V}^{2+}$  and  $\text{Cr}^{2+}$  are strong reducing agents
- Magnetic properties : Diamagnetism and paramagnetism.
- Formation of coloured ions : due to d-d transitions
- Form a large number of complex compounds
- Forms interstitial compounds : Non-stoichiometric and are neither ionic nor covalent.
- Alloy formation : Due to similar atomic sizes. (15% difference in metallic radius)

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

## Lanthanoids

- Electronic configuration:  $4f^{1-14} 5d^{0-1} 6s^2$  (Gd:  $4f^7 5d^1 6s^2$ )
- Atomic and ionic sizes: Decreases from La to Lu (Eu is the largest)
- Oxidation states: Most common is +3.
- Some elements: exhibit +2 and +4.
- General characteristics
- Silvery white soft metals and tarnish rapidly in air.
- Hardness increases with increasing atomic number.
- Metallic structure and good conductors of heat and electricity.
- Variable density
- Trivalent Lanthanoid ions are coloured.
- Ionisation Enthalpies : Low third ionisation enthalpies.
- Good reducing agents.

MISCH METAL - Alloy of Ln (95%), Fe (5%) & S, C, Ca, Al etc..

## Uses

- In production of iron and steels.
- TiO in pigment industry
- $\text{MnO}_2$  in dry battery cells.
- As catalysts in industry.
- Ni complexes: polymerization of alkynes and other organic compounds
- AgBr in photographic industry.

## Catalysts

Contact process =  $\text{V}_2\text{O}_5$

Haber process =  $\text{Fe}_2\text{O}_3 + \text{Al}_2\text{O}_3 + \text{K}_2\text{O}$

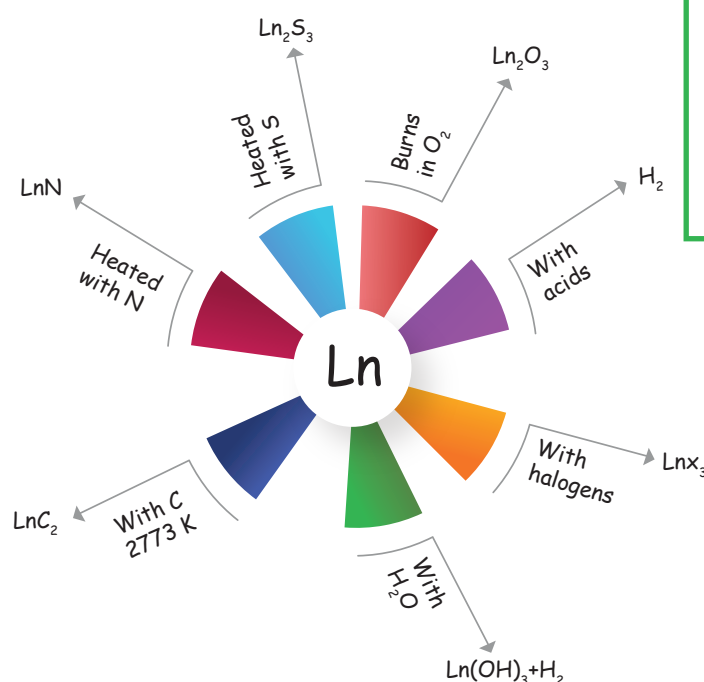
Decomposition of  $\text{KClO}_3 = \text{MnO}_2$

Ostwald process = Pt/Rh

Zeigler Natta catalyst =  $\text{TiCl}_4 + (\text{C}_2\text{H}_5)_3\text{Al}$

Hydrogenation of Alkene = Ni/Pd

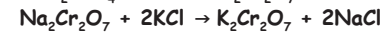
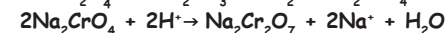
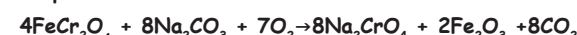
Wilkinson's catalyst =  $\text{RhCl}(\text{PPh}_3)_3$



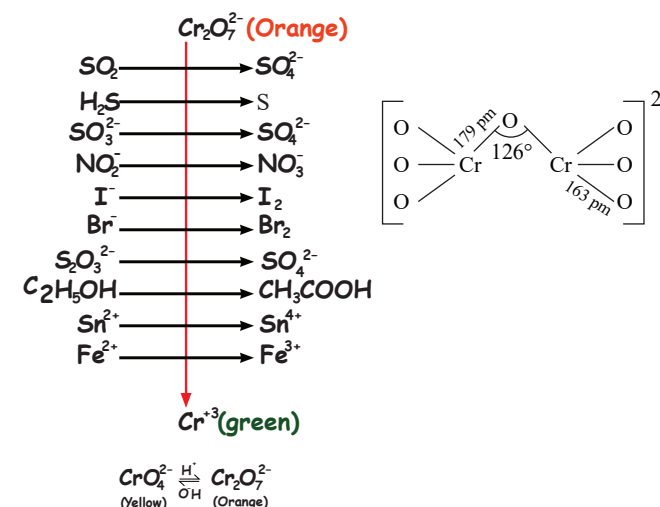
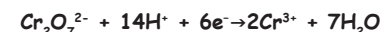
## Compounds of d-block elements

### Potassium dichromate $\text{K}_2\text{Cr}_2\text{O}_7$

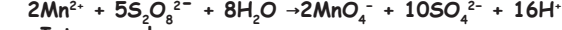
Preparation :



Properties : Strong oxidising agent



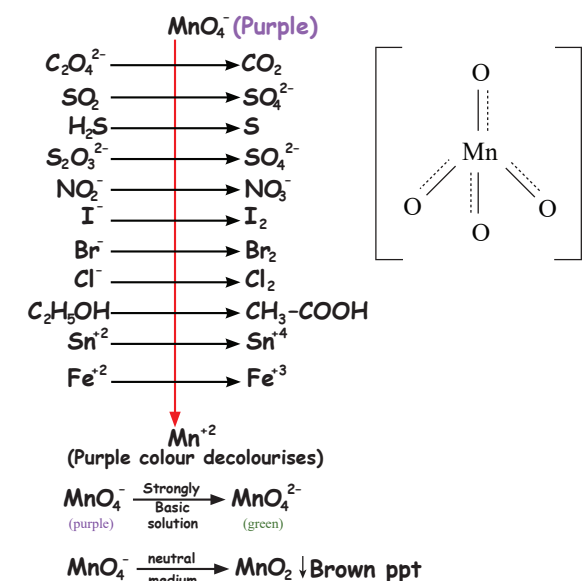
### Potassium permanganate $\text{KMnO}_4$



• Intense colour

• Paramagnetism

• Strong oxidising agent



## Actinoids

- Electronic configuration :  $[\text{Rn}]5f^{1-14} 6d^{0-1} 7s^2$
- Ionic sizes : Gradual decrease along the series
- Oxidation states : Most common is +3 . They show O.S. of +4, +5, +6 and +7.
- General characteristics :
- Highly reactive metals
- Irregularities in metallic radii, greater than in Lanthanoids.
- Magnetic properties more complex than lanthanoids
- Actinoid Contraction > Lanthanoid Contraction



PHYSICS  
WALLAH



The element that usually does not show variable oxidation state is ?

- a) Cu    b) Ti  
c) Sc    d) V

Which of the following is not formed when  $\text{H}_2\text{S}$  reacts with acidic  $\text{K}_2\text{Cr}_2\text{O}_7$  solution ?

- a)  $\text{CrSO}_4$     b)  $\text{Cr}_2(\text{SO}_4)_3$   
c)  $\text{K}_2\text{SO}_4$     d) S

d&f BLOCK ELEMENTS